Nicole Sorensen

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Education

Carnegie Mellon University, Pittsburgh, PA Expected May 2026
B.S. in Statistics and Machine Learning GPA: 3.88, Dean's List
Relevant Coursework: Causation and Machine Learning, Intro to Causal Inference, Probabilistic Graphical Models (Fall 2025), Intro to Machine Learning, Machine Learning in Practice, Probability and Statistical Inference, Data Structures and Algorithms, Multivariate Analysis, Statistical Graphics and Visualization

Research Experience

Causal Analysis of ICU Transfers and Sepsis, Carnegie Mellon University 11/2024 – 05/2025
Self-directed research project initiated in a CMU course and extended independently

- Investigated the causal impact of ICU transfers on sepsis risk using MIMIC-IV data
- Applied the Fast Causal Inference (FCI) algorithm with kernel-based conditional independence tests (KCI) to identify potential mediators such as Glasgow Coma Scale and creatinine.
- Estimated a negative average treatment effect (ATE = –0.174), suggesting ICU transfer may reduce sepsis risk, offering implications for triage protocols.

Temporal Predction of Ventilation using Deep Learning, Carnegie Mellon University 10/2024 – Present
Self-directed project launched in coursework and developed independently

- Developed a multi-input deep learning model (LSTM + ClinicalBERT) to predict ICU ventilation needs 6 hours in advance using MIMIC-IV, enabling earlier interventions
- Achieved 7× improvement over baseline in PR AUC by applying Boruta feature selection and oversampling to handle class imbalance

Diplomatic Agreements Project, Carnegie Mellon University 01/2024 – Present

- Built Python-based Scrapy, Azure AI, and Google Docs OCR pipelines to extract diplomatic agreements from various sources.
- Cleaned and organized data, contributing to a coauthored research paper with Professor John Chin.

Bridges to Healthcare Technology Program, CMU, Sponsored by Optum 06/2024 – 08/2024

- Investigated socioeconomic drivers of racial disparities in hospitalizations using County Health Rankings and Roadmaps data.
- Applied generalized additive models (GAMs) and random forests (R, scikit-learn) to identify predictive variables such as income and insurance status. Presented findings to CMU faculty and UnitedHealth Group professionals.

Professional Experience

Optum (UnitedHealth Group), Data Science Intern 05/2025 – 08/2025

- Conducted root cause analysis using LLM pipelines, k-means clustering, and analytics (Databricks, Snowflake, SQL) to identify consumer abrasion drivers.
- Analyzed tech ops tickets with clustering and topic modeling to uncover service failure patterns.

Teaching Assistant, Probability and Statistical Inference I, CMU 08/2025 – Present

- Lead recitations, hold office hours, and grade for undergraduate statistics course.
- Support students in mastering probability and statistical concepts.

Technical Skills

- **Programming:** Python (Pandas, NumPy, scikit-learn, TensorFlow, PyTorch, causal-learn), R (dplyr, ggplot2), SQL, C
- **Tools:** Databricks, Snowflake, Excel, Azure AI
- **Skills:** Causal Inference, Causal Discovery, Machine Learning, Deep Learning, Statistical Analysis, Data Visualization

Certifications

- Machine Learning with Python (IBM)
- SQL Essential Training (LinkedIn Learning)
- R for Data Science: Analysis and Visualization (LinkedIn Learning)
- Machine Learning with Python: Foundations (LinkedIn Learning)